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Solar-Driven Photoelectrochemical Upcycling of Polyimide Plastic Waste with Safe Green Hydrogen Generation

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Arbitrary disposal of plastic waste into landfills and oceans can disturb the ecological system and even challenge human survival. Centralized plastic recycling process only works for selected types of plastics (e.g., polyethylene) with limited contribution (<10%), because of high infrastructure requirement. Comparatively, photoreforming of plastic waste for commodity and fuels production is much more facile and decentralizable, and thus holds great potential to mitigate the plastic waste challenge. To this end, a fully solar powered photoelectrochemical system is developed to selectively upgrade polyimide waste (often appears in electronic waste) into valuable commodity chemicals, including succinic acid, acetic acid, and formic acid, and cogenerating green hydrogen fuels. It is also demonstrated that one of the key monomer, pyromellitic acid, and the precious metals (in electronic waste) can be fully recycled. This proof-of-concept demonstration provides a new viewpoint for designing decentralized photoelectrochemical system for simultaneous plastic waste upcycling and renewable fuel synthesis, critical for a sustainable plastic economy. Selective cleavage of benzene ring also opens a green route for other benzene ring-containing waste upgrading.

1. Introduction

With the rapid and vast consumption of plastics in modem society, plastic disposal has drawn global concern and research attention, because the disposal of plastics into landfills and oceans not only wastes resource, but also brings environmental issues that are threats to human survival.[1] To tackle these plastic issues, various strategies for plastic recycling and degradation have been proposed, but with limited implementation. Particularly, only 5% of plastics are mechanically recycled with compromised properties, ending in lower-value products.[2] Thermolysis (pyrolysis) and gasification are also proposed to derive fuels or gases, but its implementation is limited by high energy cost. Although incineration for heat recovery is also adopted, its energy-saving is insignificant comparing with other recycling processes.[3] Moreover, all these recycling strategies employ centralized process and share restrictions including: 1) large landscape, 2) intensive energy consumption, 3) burdensome waste collection and transportation, and 4) laborious pre-conditioning to remove the contaminants. Hence, advanced infrastructure is always required.[4] Nevertheless, most of plastic wastes are generated in developing areas with poor infrastructures and has limited energy access. Consequently, only very small portion (<10%) of few types of plastic wastes [polyethylene, poly (ethylene terephthalate) (PET), and polystyrene] has been recycled.[1,4] Therefore, it is urgent to develop new and efficient plastic recycling/upcycling technology that 1) employs decentralized process, 2) requires low-energy input and can renewable energies, and 3) tolerates contaminations in the waste.

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Photocatalysis offers a decentralized, energy non-intensive process for plastic waste recycling. Recently, Reisner et al. reported a photocatalytic process to transform plastic wastes such as PET and poly (lactic acid) (PLA) into valuable chemicals such as acetate and formate and cogenerate hydrogen (H_2) fuels with solar energy,[5] which achieves plastic waste elimination and renewable fuel synthesis simultaneously. Soon after, Xie's group and Sun's group introduced the photocatalytic conversion of plastics into C2 fuels with a tandem catalytic system, wherein Nb_2O_5 overoxidized plastic waste into carbon dioxide.[6] Selectively photocatalytic upgrading of plastic waste into valuable commodity chemicals is highly beneficial while challenging, deserving further study. Additionally, comparing with photocatalyst-suspension photocatalytic system,[5] photoelectrochemical (cell could separate oxidative reforming of plastic waste and reductive hydrogen evolution in space,[7] rendering the whole process more efficient and safer.[8] More importantly, high-purity hydrogen could be obtained. Hence, photoelectrochemical reforming is thought to be a more promising approach to upcycle plastic waste for valuable commodities and fuels production. One the other hand, safe hydrogen storage technology is the premise of the wider implementation of hydrogen economy, promising for energy security.[9] Formic acid (FA) has attracted tremendous research interests as a promising H_2 carrier, because of high volumetric capacity (53 g H_2 L^{-1}), low toxicity, and excellent stability at room temperature.[10] Deriving FA and green hydrogen from plastic waste with renewable energy, for example, solar energy, would be highly beneficial for sustainable development. Photoelectrochemical upcycling of plastics waste for green chemicals and hydrogen production has garnered unprecedented attention recent years. However, almost all of them localized on less restive plastic, particularly PET, with little attention paid to those obstinate but valuable plastic waste (Table S1, Supporting Information).[11]

Polyimides (PI), a typical thermoplastic, possess excellent thermal, and chemical stability,[12] and their prominent electrical insulation properties make them prevalent in electrical and electronic applications. Particularly, PI is widely used as high-temperature insulation materials and passivation layers in the fabrication of integrated circuits. With the wider deployment of flexible electronics, the production and consumption of PI is expected to increase significantly.[13] Notably, precious metals are greatly involved in the microelectronics fabrication, a large proportion of which would be inevitably adhere/integrated with PI plastic. Hence, it is critical to recycling PI and the precious metals involved. However, the outstanding thermal and chemical properties of PI makes the conventional recycling difficult and costly.[14] For example, pyrolysis of PI film waste needs to be conducted at 500–900 °C in a centralized energy-intensive process.[14b] Thus, PI waste is widely landfilled or incinerated by now, apart from seldom reports of mechanical blending and (steam-) pyrolysis treatment. Comparatively, it is promising to develop a solar-driven technique with low-energy cost to upcycle PI waste into valuable commodities or fuels and simultaneously recovery valuable metals involved.

In this study, a fully solar powered photoelectrochemical system was developed to valorize PI waste into valuable organic acids, that is, succinic acid (SA), acetic acid (HAc), and FA with simultaneous green hydrogen generation. To avoid the overoxidation of FA, titanium doped hematite (Ti-doped α-Fe_2O_3) nanorod films with well controlled band position was designed as the photoanode to selectively cleave the benzene rings. Meanwhile, the electrons were well delivered to the Ni_2P/NF cathode for efficient hydrogen generation, avoiding electron-hole recombination loss. Up to 91.6% Faradaic efficiency was obtained for PI reforming at the photoanode in the studied system. Finally, we studied the real-world applicability of the developed fully solar powered PI upgrading system with strongly fluctuating solar electricity directly on the roof of our laboratory (**Figure 1**). Self-supported bias-free operation endows our developed process with uncompetitive flexibility and sustainability, highly applicable for on-site PI treatment.

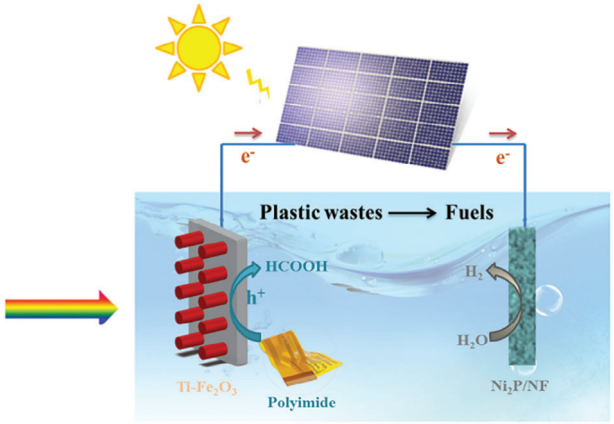

Figure 1. Schematic illustration of a solar-powered PV-PEC cell with Ti-doped Fe_2O_3 photoanode for waste PI upgrading with hydrogen cogeneration at the cathode.

2. Results and Discussion

2.1. Photoanode Fabrication

Ti-doped α-Fe_2O_3 photoanode, which was chosen because of its superior stability and high abundance, was prepared according to a modified literature procedure.[15] Briefly, Ti-doped FeOOH nanorod arrays were firstly grown on a fluorine doped tin oxide (FTO) substrate by hydrothermal reaction at 95 °C for 20 h, followed by annealing in air at 700 °C for 20 min to generate Ti-doped α-Fe_2O_3 film. As shown in **Figure 2a**, the X-ray diffraction (XRD) patterns show two characteristic diffraction peaks located at 35.6° and 64°, which are perfectly indexed to highly crystalline α-phase hematite (JCPDS 87–1165). The obtained Ti-doped α-Fe_2O_3 films exhibit visible-light absorption with optical absorption edge of about 616 nm, corresponding to an electronic bandgap of 2.01 eV (Figure 2b). Hence, high solar energy absorption is expected. Scanning electron microscopy (SEM) images show that a thin film of vertical Ti-doped α-Fe_2O_3 nanorods is uniformly grown on FTO substrate (Figure 2c), and the thickness of Ti-doped α-Fe_2O_3 layer is about 500 nm, as observed from the cross-section view image (the inset of Figure 2c). The transmission electron microscopy (TEM) image (Figure 2d) further confirms the formation of 1D Ti-doped α-Fe_2O_3 nanorod

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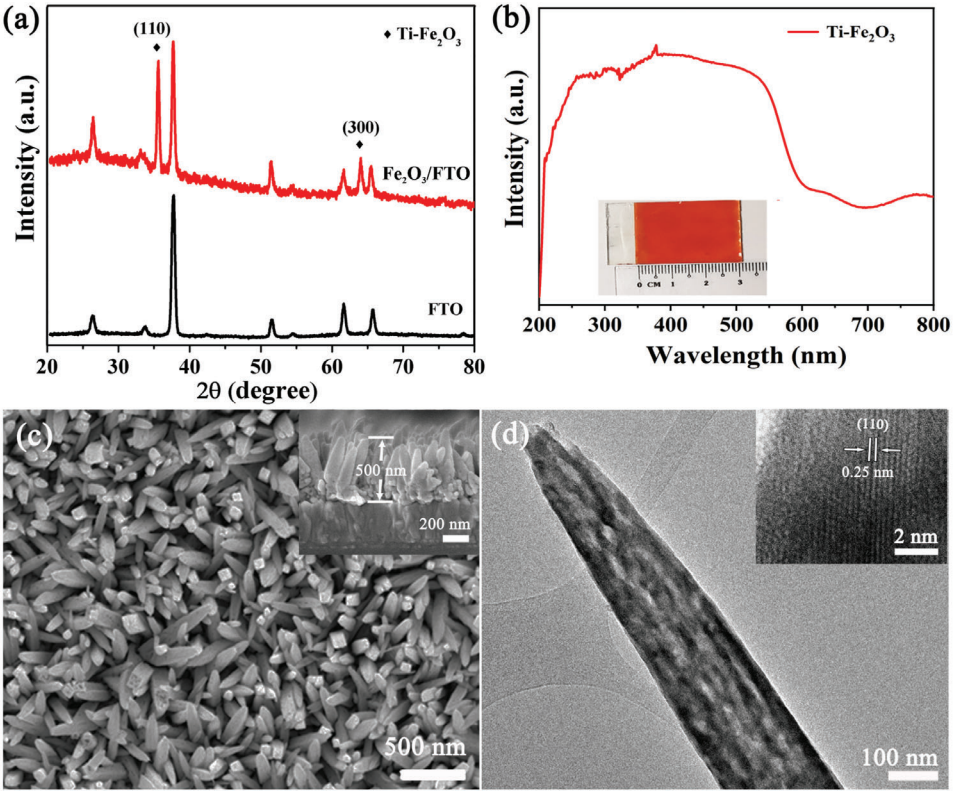

Figure 2. a) XRD patterns of pure FTO and Ti-Fe_2O_3 photoanode. b) UV-vis absorption and the photographs (the inset), c) Side-view SEM images and top-view (the inset), and d) TEM image and HR-TEM image (the inset) of Ti-doped Fe_2O_3 photoanode.

with porous nanostructure. The high-resolution image (the inset of Figure 2c) demonstrates a lattice spacing of 0.25 nm, corresponding to (110) lattice planes of α-Fe_2O_3. Moreover, the energy dispersive spectroscopy (EDS) mapping confirms the homogeneous doping of 1 at% Ti element into the α-Fe_2O_3 nanorods (Figure S1, Supporting Information), in accordance with the X-ray photoelectron spectroscopy (XPS) in Figure S2 (Supporting Information), where Ti^{4+} atoms can be well identified. In contrast to the typical binding energy of Ti^{4+} in TiO_2 (458.5 eV), a red shift of 0.4 eV was observed in our Ti-doped α-Fe_2O_3 samples. Meanwhile, the binding energy of Fe positively shifted after Ti doping (Figure S3, Supporting Information), confirming the well dispersion of Ti^{4+} into the crystal lattice of α-Fe_2O_3.

2.2. Photoelectrochemical Measurements

Photoelectrochemical performance of Ti-doped α-Fe_2O_3 photoanode was evaluated in 1.0 M KOH (pH 13.6) aqueous solution with (PI oxidation) and without (water oxidation) adding specific amount of PI under one sun illumination (AM 1.5G, 100 mW cm^{-2}). The optimum doping amount of Ti was firstly studied. As demonstrated in Figure S4 (Supporting Information), α-Fe_2O_3-20Ti exhibits the best catalytic performance for both oxygen evolution reaction (OER) and PI oxiation reaction (PIOR) among all of them. Appropriate increase of Ti^{4+} concentration contributes to the increase of electron concentration and the electrical conductivity, thus improves the charge separation significantly (reduced slope in the Mott-Schottky plot in Figure S5, Supporting Information). Furthermore, appropriate Ti doping could also enhance charge mobility and reduce charge recombination, as confirmed by the greatly reduced charge transfer resistance in Figure S6 (Supporting Information). However, too high Ti^{4+} concentration would result in an overly narrow depletion region and a subsequent reduced charge separation efficiency.[11,16] Hence, α-Fe_2O_3-25Ti exhibits decreased photoelectrochemical performance. Additionally, Ti doping results in the downshift of the flat-band position of α-Fe_2O_3, in line with the negative shift of the onset potential observed in Figure S4 (Supporting Information), which further accelerates the photoelectrochemical reaction through upward band bending. Overall, it is found that α-Fe_2O_3-20Ti shows the highest PI photoelectrochemical upgrading performance, thus it was chosen as the photoelectrochemical catalyst in the subsequent study. **Figure 3a** shows the current density-potential profiles of Ti-doped α-Fe_2O_3 photoanode under dark and illumination conditions. The photocurrent density of the Ti-doped α-Fe_2O_3 photoanodes under no PI added condition is 1.5 mA cm^{-2} at 1.23 $V_{RHE,}$ and the corresponding photo-onset potential of Ti-doped α-Fe_2O_3 photoanodes is 0.84 V_{RHE}, which is comparable to the state-of-the-art values of the reported Ti-doped α-Fe_2O_3 photoanode.[15,17] Subsequently, specific amount of 1.0 M KOH solution containing 600 mg mL^{-1} PI waste was added to study the performance for PIOR. The dissolution of PI waste in KOH solution is depicted in Methods. As shown in Figure 3a, one can observe that the photocurrent of PIOR

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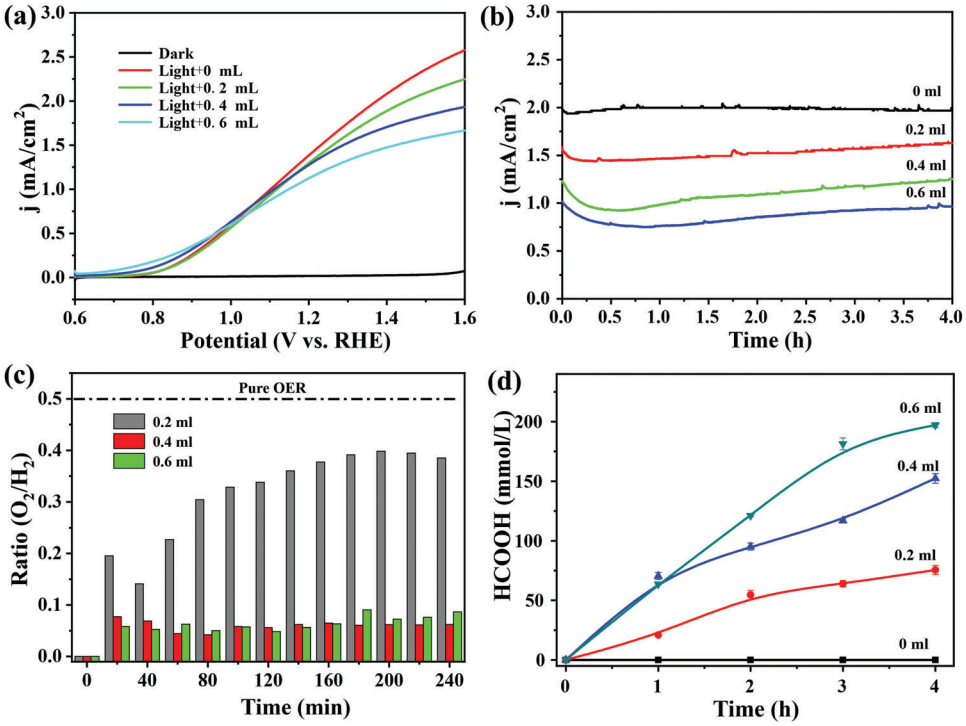

Figure 3. a) Current density-potential profiles of Ti-doped α-Fe_2O_3 photoanode in 1.0 M KOH at dark (black curve) or AM 1.5G illumination conditions (color curves) and with different amount of PI solution addition. b) Current density-time curves of Ti-doped α-Fe_2O_3 photoanode with different amount of PI in 1.0 M KOH at 1.2 V_{RHE} under AM 1.5G illumination. c) The quantified O_2/H_2 ratio and d) photoelectrochemical production rate of FA in a single-compartment reactor containing different amount of PI in 1.0 M KOH at 1.2 V_{RHE} under AM 1.5G illumination. Error bar stands for the standard deviation.

appears earlier than water oxidation (0.84 V_{RHE}), suggesting PIOR is thermodynamically more favorable than water oxidation, which was confirmed by theoretical oxidation potential calculation in Note S1 (Supporting Information). On the other hand, the gradually increased Tafel slope along with the increase of PI amount in Figure S7 (Supporting Information) indicates the poor kinetics of PIOR, identical with the large overpotential observed in Figure 3a. That is because PIOR is a multiple-electron transfer reaction and endure high steric hindrance. In contrast, negligible catalytic performance was observed for the FTO substrate (Figure S8, Supporting Information).

It is noted that the photocurrent of PIOR decreased a bit comparing with water oxidation after 1.0 V_{RHE}, which could be attributed to the following factors. One is that the light penetration was blocked by the darkening brown color of the solute of PI, which was verified by the UV–vis absorption spectra in Figure S9 (Supporting Information). Accordingly, reduced photocurrent was observed with increased PI amount in Figure 3a. The second reason is that, along with the increase of applied voltage, PI reactant adjacent to the electrode would become insufficient and water oxidation would gradually overtake (onset potential of 0.84 V_{RHE}), which would take up part of the active sites for PI oxidation. Noteworthily, water oxidation is less thermodynamically favorable than PI oxidation. Hence, slowdown of current ramping is expected. Third, the cumbersome mass transport of partially depolymerized PI molecules could slow down the reaction as well (Figures S10 and S11, Supporting Information). Collectively, photocurrent of PIOR decreases a bit after 1.0 V_{RHE}. Therefore, a carefully chosen PI concentration is needed to optimize the energy efficiency of the process.

The long-term stability of Ti-doped α-Fe_2O_3 photoanode with different amount of PI waste added was further tested at applied potential of 1.2 V_{RHE}. A voltage of 1.2 V_{RHE} was chosen herein because too low operation voltage unfavors the production of FA and demonstrates low reaction rates, while too high operation voltage would result in inevitable loss of TOC and Faradaic efficiency (Figure S12, Supporting Information). As shown in Figure 3b, the Ti-doped α-Fe_2O_3 photoanode exhibits a stable photocurrent during four consecutive 4-h test period, revealing the excellent stability of Ti-doped α-Fe_2O_3 photoanode against photo-corrosion, highly beneficial for the developed PI oxidation application. The high structural and crystalline robustness were also confirmed by SEM and XRD analysis after long term measurement (Figure S13, Supporting Information). After PI waste was added into the solution, the photocurrent slightly decreased initially due to the decreased light penetration, arising from the darkening color of the dissolved PI plastic and the gradually evolved intermediates, such as hydroquinone (HQ) and benzoquinone (BQ) (**Figure 4**; Figures S9 and S14, Supporting Information). Thereafter, the photocurrent slowly recovered with the reaction proceeds, because dark color compounds were gradually consumed that recovered light penetration, which is consistent with the increasing of O_2 evolution as the reaction proceeds (Figure 3c).

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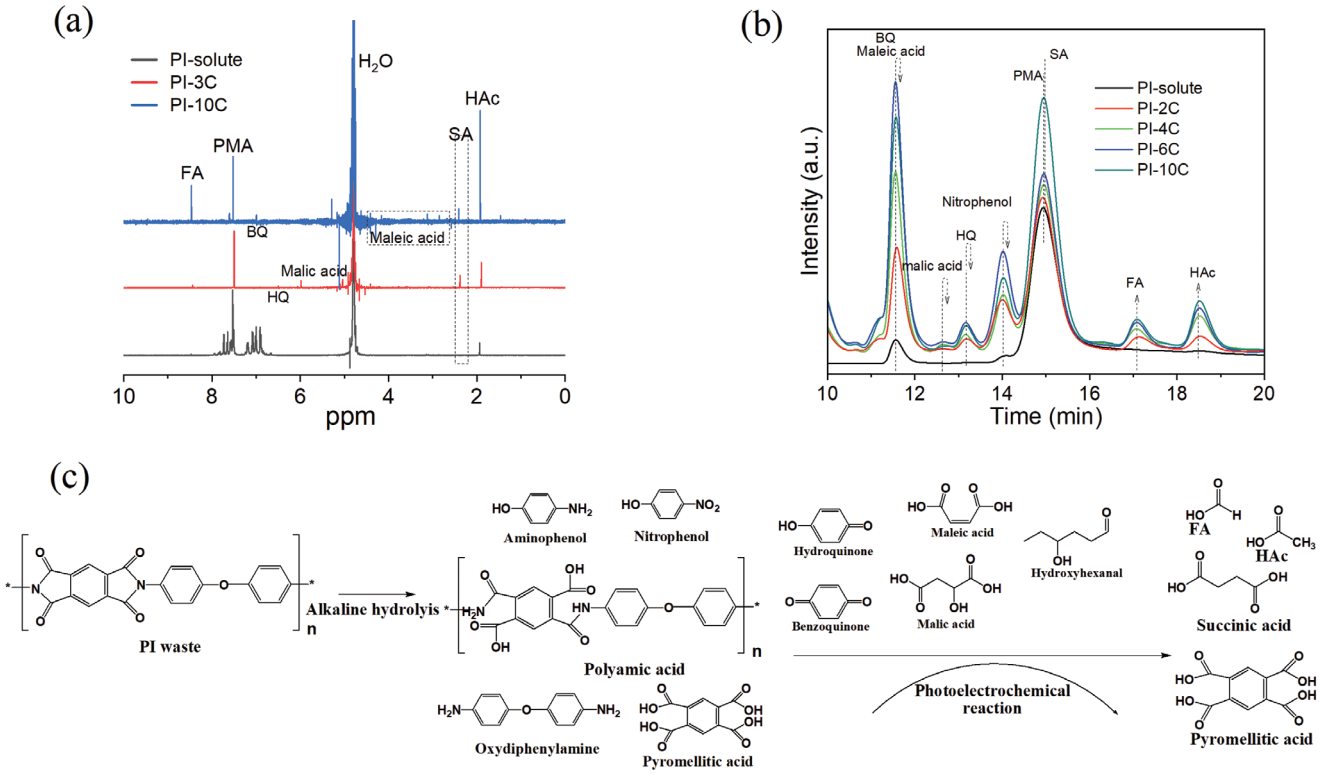

Figure 4. a) ^{1}H NMR spectra of dissolved PI waste before and after different photoelectrochemical reforming duration, showing the gradual evolution of formic acid, acetic acid, and succinic acid along with some intermediates. b) HPLC spectra of dissolved PI waste before and after different oxidation duration to show the main products and intermediates. c) Proposed PI upcycling pathways based on detailed products and intermediates analysis.

2.3. Products Analysis and Reaction Mechanism

Afterwards, detailed product analysis was performed to disclose the possible reaction mechanism. As indicated in Figures S10 and S11 (Supporting Information), incomplete depolymerization of PI was observed, in line with the small signals of 4,4′-oxydiphenylamine (ODA) and pyromellitic acid (PMA) monomers identified in the ^{1}H nuclear magnetic resonance (NMR) spectra of PI hydrolysate (Figure S15, Supporting Information). Besides, multiple new peaks evolved apart from the ODA and PMA monomers during the alkaline treatment process, indicating the possibility of the degradation of monomers. Compounds, such as aminophenol, nitrophenol, HQ, and BQ were identified through careful inspection, which explains the dark brown color observed.[18] Subsequently, photoelectrochemical reforming of PI solutes was performed, and aliquot sampling was conducted at different reaction duration to disclose the possible reaction pathways (Figures 4a,b). We found that PMA was quite stable in the photoelectrochemical reforming process, while ODA and the decomposed intermediates would go through stepwise ring-opening and oxidative bond breakings (Figures S16 and S17, Supporting Information). FA, HAc, and SA as the three major products [corroborated by gas chromatography–mass spectrum (GC–MS) in Figure S18, Supporting Information] increased gradually, while intermediates including HQ, BQ, malic acid, maleic acid, and hydroxyhexanal demonstrated initial increase and subsequent decrease during the photoelectrochemical reforming process. Besides, we also traced the reaction pathways of nitrogen with TOC/TN analyzer and ion chromatography (IC). It is found that the all the dangling -NH_2 was gradually oxidized, and nitrate was dominantly formed (over 95.7%) with the rest being released into atmosphere in the form of nitrogen gas during the photoelectrochemical conversion process, in line with the theoretic calculation in Note S1 (Supporting Information) and previous findings.[19] Accordingly, possible reaction pathways were proposed in Figure 4c. First, gradually hydrolytic cleavage of imide or amide bonds occurrs during alkaline treatment, which exposes sufficient hydrophilic groups, thus resulting in the dissolution of PI plastic, and ensuring swift charge transfer during the subsequent photoelectrochemical reforming process. Notably, the highly hydrolysis-resistant poly (amic acid) (PAA) phase unavoidably evolved during alkaline treatment renders the cleavage of imide and amide bonds incomplete.[20] Meanwhile, ODA is inevitably decomposed into aminophenol and nitrophenol. Hence, subsequent photoelectrochemical reforming was performed, wherein the —NH_2 group is gradually oxidized into —NO_2 and =O groups initially.[19,21] Because of the strong electron-withdrawing ability of the nucleophile groups (=O, —OH, or —NO_2), the electron cloud is shifted away from the benzene ring to these nucleophile groups. Subsequently, these nucleophile groups would easily lose electrons at the electron-deficient Ti doped α-Fe_2O_3 photoanode surface, resulting in the deficiency of electrons on the benzene ring and thus the weakening of bonds.[19,22] Then, the benzene ring is opened afterwards and combined with hydroxide ions, resulting in the selective production of HAc, SA, and FA.[23] The gradual conversion of ODA

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could further promote the hydrolysis of PAA phase, resulting in the complete upgrading of PI plastic, which was also confirmed by the gel permeation chromatography in Figure S11 (Supporting Information). Additionally, it is noteworthily that metals always show ultralow solubility in alkaline solution, so any precious metal involved could be effectively enriched and subject to metallurgic recovery (Figure S19, Supporting Information).

As FA is an important hydrogen carrier, hence we studied the evolution of FA in detail. As shown in Figure 3d, the concentration of FA increases with the increase of oxidation time and the amount of PI added. The amount of FA reached 197 mmol L^{-1} in the electrolyte solution at the condition of 0.6 mL PI. Noteworthily, FA is highly unstable in most reported electrooxidation system and could be overoxidized into carbon dioxide. Comparatively, our developed system shows high selectivity to FA, because of the well-chosen Ti doped α-Fe_2O_3 photoanode, which exhibits negligible oxidation ability towards FA, identical to previous reports.[11] To further disclose the reason behind, valence band position and bandgap were measured, and density functional theory (DFT) calculation was performed. As illustrated in Figures S20 and S21 (Supporting Information), all α-Fe_2O_3 samples have quite similar valence band positions and bandgaps, except α-Fe_2O_3-25Ti. That is because the doping amount of Ti is very small. On the other hand, our DFT calculation (Figure S22, Supporting Information) shows that FA exhibits much lower adsorption energy (−0.25 eV) on the prepared Ti doped α-Fe_2O_3 photoanode compared to the ODA (−0.53 eV) and other intermediates, such as HQ (−0.28 eV) and BQ (−0.96 eV). Hence, FA is stably produced in our developed system. Additionally, the O_2/H_2 ratio in gaseous products was detected using GC–thermal conductivity detector (GC-TCD) to show the selectivity of PI oxidation over water oxidation (Figure 3c). The O_2/H_2 ratio was 0.5 when no PI was added, confirming the occurring of water oxidation reaction. The O_2/H_2 ratio decreases to as low as 0.14 after adding 0.2 mL PI, suggesting PI photooxidation becomes more and more prominent. Afterward, the O_2/H_2 ratio would gradually increase with the reaction time, showcasing the continuous consumption of PI. The influence of PI amount on the O_2/H_2 ratio was also investigated. It is worth noting that the O_2/H_2 ratio decreases dramatically and remains below 0.09 with more than 0.4 mL PI added, confirming the domination of PI oxidation over water oxidation that evolves oxygen. Finally, the overall Faradaic efficiency of photoelectrochemically PI waste upcycling was calculated to be up to 91.6% with 0.4 mL PI added (Equation 1). The higher O_2/H_2 ratio observed at 20 min is ascribed to following two aspects: 1) The gradual depolymerization of PAA phase occurs during photoelectrochemical process, which accelerates the mass transport and thus the kinetics of PIOR. 2) Compared to initial ODA, the intermediates, such as HQ and BQ has smaller steric hindrance and higher adsorption strength on the Ti-doped α-Fe_2O_3 catalyst, which gradually results in the suppression of OER. Therefore, O_2 evolution reduced after 20 min. We also calculated the conversion of PI and the selectivity of the three main oxidation products, that is, FA, HAc, and SA using the Equations (2) and (3), respectively. Approximately 94.6 ± 3.3% of PI was converted each pass, and the selectivity of FA, HAc, and SA was found to be 17.9–21.4%, 45.8–50.4%, and 17.3–21.1%, respectively. Carbon balance (up to 97.5 ± 1.5%) was also verified with TOC analysis, showing insignificant carbon loss caused by overoxidation. Moreover, no significant product selectivity variance was observed under different illumination intensity, showcasing the high practicability under actual fluctuating illumination condition. Notably, undoped Fe_2O_3 demonstrated similar product selectivity, albeit with lower energy efficiency. The trace amount of Ti doping suggests that α-Fe_2O_3 serves as the active sites. In addition, the preferential adsorption of the reactant and the intermediates on the Fe sites (DFT calculation) and negligible oxygen vacancy observed (XPS spectra in Figure S2, Supporting Information) also suggest that Fe sites are the main active sites, in agreement with previous works.[24] Finally, we accessed the feasibility of the product separation and performed technoeconomic analysis with Aspen simulation (detailed in Note S2 and Figures S23–S25, Supporting Information). All the organic acids can be well separated with the process flow diagram shown in Figure S24 (Supporting Information). The levelized cost and revenue results in Figure S25 (Supporting Information) confirm that the developed PEC upgrading process is economically viable. Distillation towers and heat exchangers are the main cost units, with the heating and cooling utilities contributing to the highest energy consumption, which shows that the development of greener and less costly organic acids separation process is highly rewarding. Comparatively, the developed PEC upgrading process itself shows low energy consumption, hence it is very promising for energy reduction in plastic waste treatment. Notably, as the developed PEC process offers a green and circular solution for plastic waste treatment, potential economic benefits from avoiding carbon tax is anticipated, although this part of saving on carbon tax income was not considered herein. The highest revenue comes from the selling of PMA, as the conventional production of which is highly energy consuming and carbon intensive. We also confirmed that high purity PMA can be recovered after acidification (Figure S26, Supporting Information). Additionally, PEC upcycling of actual PI plastic wastes generated from a standard class-100 microfabrication clean room (e-beam PI wastes generated during microelectronic device fabrication) was also accessed. As depicted in Figure S19 (Supporting information), contaminated PI plastic could also be effectively upgraded into organic acids, while the metals involved can be fully recovered, highlighting practical applicability. Moreover, it is noted that the solar-to-fuel (STF) efficiency is not a decisive metric for solar-driven reforming system because the replaced organic oxidation reactions are always thermodynamically favorable than water oxidation reactions, resulting in reduced STF efficiency, and the chemical bias in the system is always challenging to be accurately measured because of multiple reaction pathways. Hence, STF efficiency does not provide an accurate assessment on the economic potential of the developed PEC waste reforming process. Nonetheless, we can notice the high light utilization efficiency in the developed PEC system based on the high Faradaic efficiency. Besides, at least 5 times higher energy efficiency than the traditional photocatalysis system was verified when compared with dispersed Ti doped α-Fe_2O_3 counterpart, in accordance with previous report.[11]

2.4. Full Solar-Driven PI Reforming with Hydrogen Co-Generation

To implement fully solar powered PI waste reforming, a commercial PV panel (Figure S27, Supporting Information) with a

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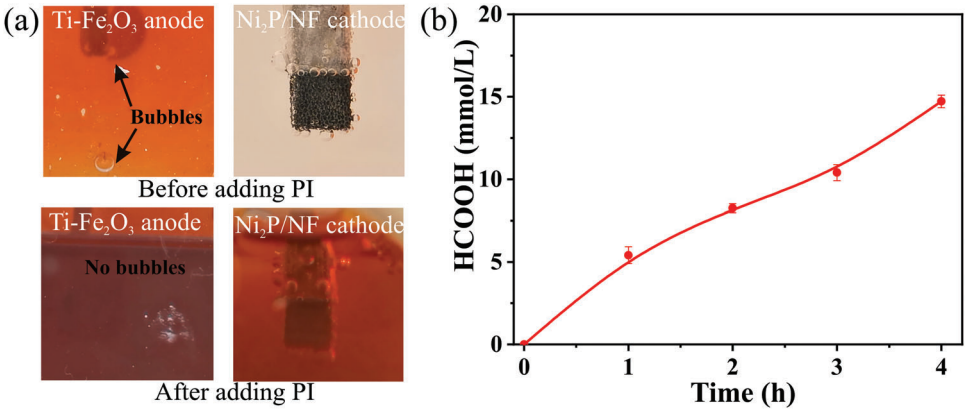

Figure 5. a) Electrode photographs for fully solar powered photoelectrochemical process with (below) and without (upper) adding of PI plastic under solar light irradiation. b) Full solar-driven photoelectrocatalytic production curve of FA. Error bar stands for the standard deviation.

power of 6 W was added, and the whole setup was driven by actual sunlight on the roof of our lab (N 1° 20′54.4″; E 103° 40′59.3″). The solar energy plays twofold roles herein: 1) to offer energetic photo-carriers at the photoanode to oxidize PI, and 2) to provide the necessary bias through PV panel. The detailed setup was shown in Figure S28 (Supporting Information), where 0.9 mL PI was added and the newly prepared Ti-doped α-Fe_2O_3 nanorod films and Ni_2P/NF were used at the photoanode and cathode, respectively. First, fully solar powered photoelectrochemical water oxidation (without adding PI) was recorded. Around 2.1 V work potential was supplied by the PV panel and ~11.33 mA current (the area of photoanode film is 6.8 cm^2) was obtained. One can clearly see the bubbling of gases on both electrodes (**Figure 5a**). Subsequently, 0.9 mL PI solution was added to study the performance of photoelectrochemical PIOR. The recorded potential on the electrolysis cell was still around 2.1 V, but the photocurrent was reduced to ~6.8 mA, which is due to the reduced light penetration by the dark color of PI solution. Hence, it is critical to engineer the setup to reduce the length of light pathway for real application in the future. Notably, comparing with the vibrant hydrogen bubbling on the cathode, the oxygen bubbling on the photoanode was greatly suppressed, confirming the domination of photooxidation of PI. A short movie was also recorded for more vivid presentation (Movie S1, Supporting Information). Finally, the amount of FA was also quantified, and the evolution plot is depicted in Figure 5b. The gradual accumulation of FA was observed and up to 14.6 mM FA was obtained within 4 h duration. Therefore, fully solar powered PI waste reforming for commodity chemicals (including FA, HAc, and SA) and green hydrogen generation is highly applicable. A detailed comparison with previously reported photoelectrochemical plastic upgrading and FA production from organic oxidation is tabulated in Table S1 (Supporting Information). Notably, our unassisted PV-PEC system exhibits outstanding plastic upgrading and FA production performance in terms of current density and FA production rate under no external bias applied.

3. Conclusion

In summary, a fully solar powered upcycling system for PI plastic waste was developed herein to produce valuable commodity chemicals (i.e., FA, HAc, and SA) and green hydrogen. Significantly, the developed system shows high selectivity to FA because of well-designed valence band position. More importantly, the PMA monomer was effectively recycled for virgin PI production and any valuable metals involved in PI waste could be well recovered. In sharp contrast to previous work, which typically centralized on those easily tackleable plastic wastes, such as PET and PLA, we realized the effective photoelectrochemical upcycling of stubborn PI plastic waste for the first time with the developed Ti-doped α-Fe_2O_3 photoanode. Overall, our work shows a proof-of-concept of an energy non-intensive and environment-friendly process that could be decentralized for plastic upcycling in developing areas with poor infrastructure and limited energy access but rich of plastic wastes, highly beneficial for sustainable development. Promisingly, the developed process is also highly applicable to other heteroatom backboned polymers, such as PET and polyurethane, deserving further investigation.

4. Experimental Section

Preparation of Ti-doped Fe_2O_3 Nanorod Thin Films: Ti-doped Fe_2O_3 nanorod thin films were grown on FTO coated glass substrates according to a modified literature procedure.[15] Typically, 20 mL aqueous solution containing 1.0 M $FeCl_3$, 1.0 M $NaNO_3$, and 20 µL 10 wt% $TiCl_3$ in 20–30% HCl solution was added into a Teflon vial of 25 mL capacity. Then, a piece of FTO glass cleaned with deionized (DI) water, acetone, and ethanol was obliquely placed into Teflon vial and subsequently heated in a regular oven at 95 °C for 20 h. Finally, the obtained thin films were washed with deionized water and annealed in air at 700 °C for 20 min. To synthesize pure Fe_2O_3 or different amount of Ti doped Fe_2O_3 photoanode, different volume of $TiCl_3$ solution was added accordingly, and the corresponding photoanode was labeled as α-Fe_2O_3-xTi, where x stands for the microliter of $TiCl_3$ added.

PI Waste Dissolution in KOH Solution: The Adhesive on PI waste was firstly removed by 3 M Adhesive Remover and then dissolved into 1.0 M KOH at 60 °C. After cooling to room temperature, the solution was centrifuged to remove any metals precipitated and stored in a 4 °C refrigerator prior to test, and the resulted concentration of PI in KOH solution for photoelectrochemical conversion is around 60 mg mL^{-1}.

Material Characterizations: The morphologies and thickness of the hematite nanorod thin films were characterized using a field emission scanning electron microscopy (FESEM, JEOL JSM-7600F). The crystal phase of the as-prepared samples was characterized by powder X-ray

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diffraction (XRD, Bruker D2 Phaser) equipped with Cu Kα radiation. The chemical state of Ti was investigated by X-ray photoelectron spectroscopy (XPS, Omicron EA125) and the binding energy was calibrated by C1s (284.6 eV). The selected area electron diffraction pattern and lattice structure were analyzed by a transmission electron microscope (TEM, JEOL JEM-2010F). The UV–visible absorption spectra were obtained on a UV–vis–NIR Lambda 950 spectrophotometer (Shimadazu) at room temperature. Carbon and nitrogen balance quantification was performed with TOC/TN analyzer (Shimazu, Japan). Mass identification of the products was carried out on an Agilent 8890 GC equipped with 7010B triple quad mass selective detector (GC-QQQ). Samples were injected with split ratio of 20:1 at the temperature of 280 °C and Agilent 122–5562UI (60 m × 250 µm × 0.25 µm) capillary column was used for separation. Helium was used as the carrier gas with a flow rate of 1 mL min^{-1}. The temperature program was as follows: holding at 50 °C for 3 min, then sequentially ramping to 150 °C at a rate of 10 °C min^{-1}, rampling to 220 °C at a rate of 5 °C min^{-1}, and ramping to 300 °C at rate of 10 °C min^{-1}, and holding for 10 min. The MS was operated in electron ionization (EI) mode at 70 eV, and m/z range of 45–550 Da was acquired. Prior to mass measurement, the reaction solution was first extracted with ethyl acetate and derivatized with N,O-Bis (trimethylsilyl) trifluoroacetamide (BSTFA) at the temperature of 60 °C for 30 min.

Photoelectrochemical Measurement: Photoelectrochemical measurements were performed on a Gamry workstation (Gamry Interface 1000; Gamry, PA, USA) using a three-electrode setup with a single-chamber 5 cm × 5 cm × 5 cm quartz reactor (shown in Figure S19, Supporting Information), wherein Ti-doped Fe_2O_3 nanorod electrode, Pt mesh and Ag/AgCl (sat. KCl) were used as the working, counter, and reference electrodes, respectively. An AM1.5G solar simulator (LCS-100, Newport Co., Ltd.) with power intensity of 100 mW cm^{-2} was used as the light source. The photocurrent measurement was performed in 1.0 M KOH aqueous (pH 13.6) solution at a scan rate of 10 mV s^{-1} with the different amount of PI. In case of gas product analysis, the cell was well sealed and purged with argon gas, and a gas sampling bag was connected to collect gases. All the gaseous products were measured with GC-TCD (Agilent 6890N, TCD detector, argon as carrier gas, TDX-01 packed column), and liquid products were analyzed by high-performance liquid chromatography (Agilent HPLC1260, PL Hi-Plex H column, variable wavelength detector VWD 210 nm) and ^{1}H NMR spectra (Bruker Avance III 400). All the measured potentials versus Ag/AgCl were converted to a reversible hydrogen electrode (RHE) scale according to the Nernst equation ($E_{RHE} = E_{Ag/AgCl} + 0.059\ pH + 0.197$).

Additionally, the Faradaic efficiency of PI oxidation, conversion of PI, selectivity of each oxidation product is calculated by the follow equations:

$$Faradaic\ efficienty_{\ PI\ oxidation} = \left(1 - \frac{amount_{H2}/amount_{O2}}{0.5}\right) \times 100\% \quad (1)$$

where $amount_{H2}$ and $amount_{O2}$ are the measured amount of hydrogen and oxygen at specific conditions.

$$Conversion_{PI} = \left(1 - \frac{4,4' - oxydiphenylamine\ mole\ conc._{end}}{PI\ mass\ conc._{initial}/382}\right) \times 100\% \quad (2)$$

$$Selectivity_{product} = \left(\frac{n \times mole\ of\ product}{12 \times mole\ of\ 4,4' - oxydiphenyla\min e_{converted}}\right) \times 100\% \quad (3)$$

where n is the carbon number in the product.

Besides, the yield of PMA monomer is calculated with Equation (4) below:

$$Yield_{PMA} = \left(\frac{PMAmass_{collected}/254}{PI\ mass_{added}/382}\right) \times 100\% \quad (4)$$

Photocatalysis Control Experiment: Identical preparation procedures for Ti-doped Fe_2O_3 nanorod thin films was adopted to synthesize Ti-doped Fe_2O_3 nanorod powder catalysts, except without adding FTO substrate. Subsequently, the photocatalysis control was performed with the same quartz photoreactor, but without adding of external bias. Finally, products were quantified periodically, and the yields of organic acids were calculated.

Full Solar Powered Photoelectrochemical Reforming of PI Waste Demonstration: A single chamber photoelectrochemical cell was connected to a commercial PV panel (power of 6 W monocrystalline silicon from Zibo Guosheng Solar Power Company with typical characteristic curve shown in Figure S18, Supporting Information) through a slide rheostat (see Figure S19 and Movie S1, Supporting Information, for details). For practical applications, the slide rheostat could be replaced with a rechargeable battery to store the peak current and power the system at low illumination condition to realize the continuity of the system. Newly prepared Ti-doped α-Fe_2O_3 nanorod film and Ni_2P/NF electrodes were used as the photoanode and cathode, respectively. The working current and voltage were recorded with a voltmeter and multimeter (ampere mode), respectively. The experiment was conducted on the roof of our laboratory (N 1° 20′54.4″; E 103° 40′59.3″) at 12:00–16:00 on 8th July 2023.

DFT Calculations: The DFT calculations were conducted to compare the adsorption energy of organic molecules on the hematite (α-Fe_2O_3) catalyst surface. The four-layer α-Fe_2O_3 (0001) surface constructed from its bulk structure is selected to establish the slab models since it is the most common surface. To mimic Ti-doped α-Fe_2O_3 catalysts, one Fe atom in the topmost α-Fe_2O_3 layer is replaced by a Ti atom to form α-Fe_2O_3-Ti. Successively, to compare the adsorption ability of different organic molecules on the Ti-doped catalyst surface, FA (HCOOH), ODA ($C_{12}H_{12}N_2O$), HQ ($C_6H_6O_2$), and BQ ($C_6H_4O_2$) were arranged on the α-Fe_2O_3-Ti surface, respectively. In all slab models, 3D periodicities are used, and the vacuum layers more than 15 Å perpendicular to the catalyst surface were added.

The spin-polarized DFT calculations were performed with the Vienna Ab-initio Simulation Package (VASP).[25] The project augmented wave (PAW) method was used to describe the ion-electron potential.[26] The generalized gradient approximation with the Perdew-Burke-Ernzerhof (PBE) functional was used as the exchange-correlation function.[27] The Gamma-centered 4 × 4 × 1 mesh was used for k-point sampling. The long-range dispersion interactions between adsorbed molecules and catalyst surfaces were treated by applying the DFT-D3 method of Grimme.[28] The DFT + U method was applied for the d-electrons of Fe atoms (U–J = 4.3 eV)[29] and Ti atoms (U–J = 4.0 eV).[30] The energy cutoff for the plane wave basis sets of the valence electrons were set to be 500 eV. During the geometry optimization, the bottom two layers of α-Fe_2O_3 (0001) were fixed to restore the bulk property, and other atoms were allowed to relax until the maximum force on the atoms is below 0.03 eV Å^{-1}. The energy convergence criterion in the electronic self-consistent iteration was set to 10^{-6} eV.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords

Fe_2O_3 nanorod, formic acid, metal recovery, photoelectrochemical oxidation, plastic upcycling

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